PG No. 23 C_{6h} $6/m$ [hexagonal] (internal axial quadrupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(2,2)}[g](A_u)$

** symmetry

1

** expression

$$-\frac{\sqrt{5}G_u\left(x^2+y^2-2z^2\right)}{10}+\frac{\sqrt{15}G_v\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{15}G_{xy}xy}{5}+\frac{\sqrt{15}G_{xz}xz}{5}+\frac{\sqrt{15}G_{yz}yz}{5}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(2,0)}[g](A_g)$

** symmetry

z

** expression

$$\frac{\sqrt{10}G_uz}{5}+\frac{\sqrt{30}G_{xz}x}{10}+\frac{\sqrt{30}G_{yz}y}{10}$$

$\vec{\mathbb{G}}_1^{(2,2)}[g](A_g)$

** symmetry

z

** expression

$$-\frac{3\sqrt{35}G_uz\left(3x^2+3y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_vz\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_{xy}xyz}{7}-\frac{\sqrt{105}G_{xz}x\left(x^2+y^2-4z^2\right)}{35}-\frac{\sqrt{105}G_{yz}y\left(x^2+y^2-4z^2\right)}{35}$$

$\vec{\mathbb{G}}_{1,1}^{(2,0)}[g](E_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(2,0)}[g](E_{1g})$

** symmetry

x

y

** expression

$$-\frac{\sqrt{10}G_ux}{10}+\frac{\sqrt{30}G_vx}{10}+\frac{\sqrt{30}G_{xy}y}{10}+\frac{\sqrt{30}G_{xz}z}{10}$$

$$-\frac{\sqrt{10}G_uy}{10}-\frac{\sqrt{30}G_vy}{10}+\frac{\sqrt{30}G_{xy}x}{10}+\frac{\sqrt{30}G_{yz}z}{10}$$

$\vec{\mathbb{G}}_{1,1}^{(2,2)}[g](E_{1g})$, $\vec{\mathbb{G}}_{1,2}^{(2,2)}[g](E_{1g})$

** symmetry

x

y

** expression

$$-\frac{3\sqrt{35}G_ux\left(x^2+y^2-4z^2\right)}{70}+\frac{\sqrt{105}G_vx\left(3x^2-7y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_{xy}y\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{xz}z\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{yz}xyz}{7}$$

$$-\frac{3\sqrt{35}G_uy\left(x^2+y^2-4z^2\right)}{70}+\frac{\sqrt{105}G_vy\left(7x^2-3y^2+2z^2\right)}{70}-\frac{\sqrt{105}G_{xy}x\left(x^2-4y^2+z^2\right)}{35}+\frac{\sqrt{105}G_{xz}xyz}{7}-\frac{\sqrt{105}G_{yz}z\left(x^2-4y^2+z^2\right)}{35}$$

$\vec{\mathbb{Q}}_1^{(2,1)}[g](A_u)$

** symmetry

z

** expression

$$\frac{\sqrt{30}G_vxy}{5}-\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{30}G_{xz}yz}{10}-\frac{\sqrt{30}G_{yz}xz}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(2,1)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{1,2}^{(2,1)}[g](E_{1u})$

** symmetry

x

y

** expression

$$-\frac{3\sqrt{10}G_u yz}{10} - \frac{\sqrt{30}G_v yz}{10} + \frac{\sqrt{30}G_{xy} xz}{10} - \frac{\sqrt{30}G_{xz} xy}{10} - \frac{\sqrt{30}G_{yz}\,(y-z)\,(y+z)}{10}$$

$$\frac{3\sqrt{10}G_u xz}{10} - \frac{\sqrt{30}G_v xz}{10} - \frac{\sqrt{30}G_{xy} yz}{10} + \frac{\sqrt{30}G_{xz}\,(x-z)\,(x+z)}{10} + \frac{\sqrt{30}G_{yz} xy}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{2}G_{xz} y}{2} - \frac{\sqrt{2}G_{yz} x}{2}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](A_g)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{5\sqrt{42}G_v xyz}{14} - \frac{5\sqrt{42}G_{xy} z\,(x-y)\,(x+y)}{28} - \frac{\sqrt{42}G_{xz} y\left(x^2+y^2-4z^2\right)}{28} + \frac{\sqrt{42}G_{yz} x\left(x^2+y^2-4z^2\right)}{28}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,-1)}[g](E_{1g})$, $\vec{\mathbb{Q}}_{2,2}^{(2,-1)}[g](E_{1g})$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}G_u x}{2} + \frac{\sqrt{6}G_v x}{6} + \frac{\sqrt{6}G_{xy} y}{6} - \frac{\sqrt{6}G_{xz} z}{6}$$

$$\frac{\sqrt{2}G_u y}{2} - \frac{\sqrt{6}G_v y}{6} + \frac{\sqrt{6}G_{xy} x}{6} - \frac{\sqrt{6}G_{yz} z}{6}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,1)}[g](E_{1g})$, $\vec{\mathbb{Q}}_{2,2}^{(2,1)}[g](E_{1g})$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$-\frac{\sqrt{42}G_u x\left(x^2+y^2-4z^2\right)}{28} - \frac{\sqrt{14}G_v x\left(x^2-9y^2+6z^2\right)}{28} - \frac{\sqrt{14}G_{xy} y\left(3x^2-2y^2+3z^2\right)}{14} + \frac{\sqrt{14}G_{xz} z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$-\frac{\sqrt{42}G_u y\left(x^2+y^2-4z^2\right)}{28} - \frac{\sqrt{14}G_v y\left(9x^2-y^2-6z^2\right)}{28} + \frac{\sqrt{14}G_{xy} x\left(2x^2-3y^2-3z^2\right)}{14} + \frac{\sqrt{14}G_{yz} z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,-1)}[g](E_{2g})$, $\vec{\mathbb{Q}}_{2,2}^{(2,-1)}[g](E_{2g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$\frac{\sqrt{6}G_{xy} z}{3} - \frac{\sqrt{6}G_{xz} y}{6} - \frac{\sqrt{6}G_{yz} x}{6}$$

$$\frac{\sqrt{6}G_v z}{3} - \frac{\sqrt{6}G_{xz} x}{6} + \frac{\sqrt{6}G_{yz} y}{6}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,1)}[g](E_{2g})$, $\vec{\mathbb{Q}}_{2,2}^{(2,1)}[g](E_{2g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$-\frac{5\sqrt{42}G_u xyz}{14} + \frac{\sqrt{14}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{28} - \frac{\sqrt{14}G_{xz}y\left(9x^2-y^2-6z^2\right)}{28} + \frac{\sqrt{14}G_{yz}x\left(x^2-9y^2+6z^2\right)}{28}$$

$$-\frac{5\sqrt{42}G_u z\left(x-y\right)\left(x+y\right)}{28} + \frac{\sqrt{14}G_v z\left(3x^2+3y^2-2z^2\right)}{28} - \frac{\sqrt{14}G_{xz}x\left(2x^2-3y^2-3z^2\right)}{14} - \frac{\sqrt{14}G_{yz}y\left(3x^2-2y^2+3z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](A_u)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$G_u$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](A_u)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{14}G_u\left(x^2+y^2-2z^2\right)}{14} - \frac{\sqrt{42}G_v\left(x-y\right)\left(x+y\right)}{14} - \frac{\sqrt{42}G_{xy}xy}{7} + \frac{\sqrt{42}G_{xz}xz}{14} + \frac{\sqrt{42}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](A_u)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{14}G_u\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{56} - \frac{5\sqrt{42}G_v\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}$$
$$- \frac{5\sqrt{42}G_{xy}xy\left(x^2+y^2-6z^2\right)}{84} - \frac{5\sqrt{42}G_{xz}xz\left(3x^2+3y^2-4z^2\right)}{84} - \frac{5\sqrt{42}G_{yz}yz\left(3x^2+3y^2-4z^2\right)}{84}$$

$\vec{\mathbb{G}}_{2,1}^{(2,-2)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(2,-2)}[g](E_{1u})$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$G_{yz}$$

$$-G_{xz}$$

$\vec{\mathbb{G}}_{2,1}^{(2,0)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(2,0)}[g](E_{1u})$

** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{42}G_u yz}{14} - \frac{3\sqrt{14}G_v yz}{14} + \frac{3\sqrt{14}G_{xy}xz}{14} + \frac{3\sqrt{14}G_{xz}xy}{14} - \frac{\sqrt{14}G_{yz}\left(2x^2-y^2-z^2\right)}{14}$$

$$-\frac{\sqrt{42}G_u xz}{14} - \frac{3\sqrt{14}G_v xz}{14} - \frac{3\sqrt{14}G_{xy}yz}{14} - \frac{\sqrt{14}G_{xz}\left(x^2-2y^2+z^2\right)}{14} - \frac{3\sqrt{14}G_{yz}xy}{14}$$

$\vec{\mathbb{G}}_{2,1}^{(2,2)}[g](E_{1u})$, $\vec{\mathbb{G}}_{2,2}^{(2,2)}[g](E_{1u})$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$-\frac{5\sqrt{42}G_u yz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{14}G_v yz\left(9x^2-5y^2+2z^2\right)}{84}-\frac{5\sqrt{14}G_{xy}xz\left(x^2-6y^2+z^2\right)}{42}$$
$$-\frac{5\sqrt{14}G_{xz}xy\left(x^2+y^2-6z^2\right)}{42}+\frac{\sqrt{14}G_{yz}\left(x^4-3x^2y^2-3x^2z^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$$\frac{5\sqrt{42}G_u xz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{14}G_v xz\left(5x^2-9y^2-2z^2\right)}{84}-\frac{5\sqrt{14}G_{xy}yz\left(6x^2-y^2-z^2\right)}{42}$$
$$+\frac{\sqrt{14}G_{xz}\left(4x^4+3x^2y^2-27x^2z^2-y^4+3y^2z^2+4z^4\right)}{42}+\frac{5\sqrt{14}G_{yz}xy\left(x^2+y^2-6z^2\right)}{42}$$

$\vec{\mathbb{G}}_{2,1}^{(2,-2)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(2,-2)}[g](E_{2u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$G_v$$

$$-G_{xy}$$

$\vec{\mathbb{G}}_{2,1}^{(2,0)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(2,0)}[g](E_{2u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_u\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{14}G_v\left(x^2+y^2-2z^2\right)}{14}+\frac{3\sqrt{14}G_{xz}xz}{14}-\frac{3\sqrt{14}G_{yz}yz}{14}$$

$$\frac{\sqrt{42}G_u xy}{7}-\frac{\sqrt{14}G_{xy}\left(x^2+y^2-2z^2\right)}{14}-\frac{3\sqrt{14}G_{xz}yz}{14}-\frac{3\sqrt{14}G_{yz}xz}{14}$$

$\vec{\mathbb{G}}_{2,1}^{(2,2)}[g](E_{2u})$, $\vec{\mathbb{G}}_{2,2}^{(2,2)}[g](E_{2u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

$$-\sqrt{3}xy$$

** expression

$$-\frac{5\sqrt{42}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}+\frac{\sqrt{14}G_v\left(19x^4-102x^2y^2-12x^2z^2+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$+\frac{5\sqrt{14}G_{xy}xy\left(x-y\right)\left(x+y\right)}{12}+\frac{5\sqrt{14}G_{xz}xz\left(5x^2-9y^2-2z^2\right)}{84}+\frac{5\sqrt{14}G_{yz}yz\left(9x^2-5y^2+2z^2\right)}{84}$$

$$\frac{5\sqrt{42}G_u xy\left(x^2+y^2-6z^2\right)}{84}-\frac{5\sqrt{14}G_v xy\left(x-y\right)\left(x+y\right)}{12}+\frac{\sqrt{14}G_{xy}\left(4x^4-27x^2y^2+3x^2z^2+4y^4+3y^2z^2-z^4\right)}{42}$$
$$-\frac{5\sqrt{14}G_{xz}yz\left(6x^2-y^2-z^2\right)}{42}+\frac{5\sqrt{14}G_{yz}xz\left(x^2-6y^2+z^2\right)}{42}$$

* Harmonics for rank 3
$\vec{\mathbb{G}}_{3}^{(2,-2)}[g](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}G_uz}{5}-\frac{\sqrt{5}G_{xz}x}{5}-\frac{\sqrt{5}G_{yz}y}{5}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](A_g)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_uz\left(3x^2+3y^2-2z^2\right)}{15}-\frac{\sqrt{5}G_vz\left(x-y\right)\left(x+y\right)}{2}-\sqrt{5}G_{xy}xyz-\frac{\sqrt{5}G_{xz}x\left(x^2+y^2-4z^2\right)}{20}-\frac{\sqrt{5}G_{yz}y\left(x^2+y^2-4z^2\right)}{20}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](A_g)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{330}G_uz\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}-\frac{7\sqrt{110}G_vz\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}-\frac{7\sqrt{110}G_{xy}xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{110}G_{xz}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}+\frac{\sqrt{110}G_{yz}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_g,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{2}G_vy}{2}+\frac{\sqrt{2}G_{xy}x}{2}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_g,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{2}G_vx}{2}-\frac{\sqrt{2}G_{xy}y}{2}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_g,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{5\sqrt{6}G_uy\left(3x^2-y^2\right)}{24}+\frac{\sqrt{2}G_vy\left(x^2+y^2-4z^2\right)}{8}+\frac{\sqrt{2}G_{xy}x\left(x^2+y^2-4z^2\right)}{8}+\frac{5\sqrt{2}G_{xz}xyz}{4}+\frac{5\sqrt{2}G_{yz}z\left(x-y\right)\left(x+y\right)}{8}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_g,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{5\sqrt{6}G_ux\left(x^2-3y^2\right)}{24}+\frac{\sqrt{2}G_vx\left(x^2+y^2-4z^2\right)}{8}-\frac{\sqrt{2}G_{xy}y\left(x^2+y^2-4z^2\right)}{8}+\frac{5\sqrt{2}G_{xz}z\left(x-y\right)\left(x+y\right)}{8}-\frac{5\sqrt{2}G_{yz}xyz}{4}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_g,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{7\sqrt{33}G_u y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{264}+\frac{\sqrt{11}G_v y\left(53x^4-104x^2y^2-6x^2z^2+11y^4-6y^2z^2+4z^4\right)}{88}$$
$$-\frac{\sqrt{11}G_{xy}x\left(5x^4-53x^2y^2+3x^2z^2+26y^4+3y^2z^2-2z^4\right)}{44}+\frac{7\sqrt{11}G_{xz}xyz\left(7x^2-5y^2-2z^2\right)}{44}-\frac{7\sqrt{11}G_{yz}z\left(x^4-9x^2y^2+x^2z^2+2y^4-y^2z^2\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_g,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{7\sqrt{33}G_u x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{264}+\frac{\sqrt{11}G_v x\left(11x^4-104x^2y^2-6x^2z^2+53y^4-6y^2z^2+4z^4\right)}{88}$$
$$+\frac{\sqrt{11}G_{xy}y\left(26x^4-53x^2y^2+3x^2z^2+5y^4+3y^2z^2-2z^4\right)}{44}+\frac{7\sqrt{11}G_{xz}z\left(2x^4-9x^2y^2-x^2z^2+y^4+y^2z^2\right)}{44}+\frac{7\sqrt{11}G_{yz}xyz\left(5x^2-7y^2+2z^2\right)}{44}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](E_{1g})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$$

** expression

$$\frac{\sqrt{10}G_u x}{5}-\frac{\sqrt{30}G_v x}{30}-\frac{\sqrt{30}G_{xy}y}{30}+\frac{2\sqrt{30}G_{xz}z}{15}$$

$$\frac{\sqrt{10}G_u y}{5}+\frac{\sqrt{30}G_v y}{30}-\frac{\sqrt{30}G_{xy}x}{30}+\frac{2\sqrt{30}G_{yz}z}{15}$$

$\vec{\mathbb{G}}_{3,1}^{(2,0)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(2,0)}[g](E_{1g})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$$

** expression

$$-\frac{3\sqrt{10}G_u x\left(x^2+y^2-4z^2\right)}{40}-\frac{\sqrt{30}G_v x\left(11x^2-9y^2-24z^2\right)}{120}-\frac{\sqrt{30}G_{xy}y\left(21x^2+y^2-24z^2\right)}{120}+\frac{\sqrt{30}G_{xz}z\left(9x^2-21y^2+4z^2\right)}{120}+\frac{\sqrt{30}G_{yz}xyz}{4}$$

$$-\frac{3\sqrt{10}G_u y\left(x^2+y^2-4z^2\right)}{40}-\frac{\sqrt{30}G_v y\left(9x^2-11y^2+24z^2\right)}{120}-\frac{\sqrt{30}G_{xy}x\left(x^2+21y^2-24z^2\right)}{120}+\frac{\sqrt{30}G_{xz}xyz}{4}-\frac{\sqrt{30}G_{yz}z\left(21x^2-9y^2-4z^2\right)}{120}$$

$\vec{\mathbb{G}}_{3,1}^{(2,2)}[g](E_{1g})$, $\vec{\mathbb{G}}_{3,2}^{(2,2)}[g](E_{1g})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4}-\frac{\sqrt{6}xy^2}{4}+\sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4}-\frac{\sqrt{6}y^3}{4}+\sqrt{6}yz^2$$

** expression

$$\frac{3\sqrt{55}G_u x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{88}-\frac{\sqrt{165}G_v x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{264}$$
$$-\frac{\sqrt{165}G_{xy}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{132}$$
$$-\frac{\sqrt{165}G_{xz}z\left(18x^4+15x^2y^2-41x^2z^2-3y^4+y^2z^2+4z^4\right)}{132}-\frac{7\sqrt{165}G_{yz}xyz\left(x^2+y^2-2z^2\right)}{44}$$

$$\frac{3\sqrt{55}G_u y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{88}-\frac{\sqrt{165}G_v y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{264}$$
$$+\frac{\sqrt{165}G_{xy}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{132}-\frac{7\sqrt{165}G_{xz}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{\sqrt{165}G_{yz}z\left(3x^4-15x^2y^2-x^2z^2-18y^4+41y^2z^2-4z^4\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](E_{2g})$
** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_{xy}z}{3} + \frac{\sqrt{3}G_{xz}y}{3} + \frac{\sqrt{3}G_{yz}x}{3}$$

$$\frac{\sqrt{3}G_{v}z}{3} + \frac{\sqrt{3}G_{xz}x}{3} - \frac{\sqrt{3}G_{yz}y}{3}$$

$\vec{\mathbb{G}}_{3,1}^{(2,0)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(2,0)}[g](E_{2g})$
** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{6} + \frac{\sqrt{3}G_{xz}y\left(3x^2-2y^2+3z^2\right)}{6} - \frac{\sqrt{3}G_{yz}x\left(2x^2-3y^2-3z^2\right)}{6}$$

$$\frac{\sqrt{3}G_{v}z\left(3x^2+3y^2-2z^2\right)}{6} + \frac{\sqrt{3}G_{xz}x\left(x^2-9y^2+6z^2\right)}{12} + \frac{\sqrt{3}G_{yz}y\left(9x^2-y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(2,2)}[g](E_{2g})$, $\vec{\mathbb{G}}_{3,2}^{(2,2)}[g](E_{2g})$
** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{22}G_{u}xyz\left(x^2+y^2-2z^2\right)}{44} + \frac{21\sqrt{66}G_{v}xyz\left(x-y\right)\left(x+y\right)}{44} - \frac{\sqrt{66}G_{xy}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{66}$$
$$-\frac{\sqrt{66}G_{xz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{66} + \frac{\sqrt{66}G_{yz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{66}$$

$$-\frac{21\sqrt{22}G_{u}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88} + \frac{\sqrt{66}G_{v}z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{264} + \frac{21\sqrt{66}G_{xy}xyz\left(x-y\right)\left(x+y\right)}{44}$$
$$-\frac{\sqrt{66}G_{xz}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{132} - \frac{\sqrt{66}G_{yz}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{132}$$

$\vec{\mathbb{Q}}_{3}^{(2,-1)}[g](A_u)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{30}G_{v}xy}{10} + \frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{20} + \frac{\sqrt{30}G_{xz}yz}{5} - \frac{\sqrt{30}G_{yz}xz}{5}$$

$\vec{\mathbb{Q}}_{3}^{(2,1)}[g](A_u)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{15}G_{v}xy\left(x^2+y^2-6z^2\right)}{6} + \frac{\sqrt{15}G_{xy}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12} - \frac{\sqrt{15}G_{xz}yz\left(3x^2+3y^2-4z^2\right)}{12} + \frac{\sqrt{15}G_{yz}xz\left(3x^2+3y^2-4z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3}^{(2,-1)}[g](B_u,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{3}G_v xz}{2} + \frac{\sqrt{3}G_{xy}yz}{2} + \frac{\sqrt{3}G_{xz}\left(x-y\right)\left(x+y\right)}{4} - \frac{\sqrt{3}G_{yz}xy}{2}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](B_u, 2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{3}G_v yz}{2} + \frac{\sqrt{3}G_{xy}xz}{2} - \frac{\sqrt{3}G_{xz}xy}{2} - \frac{\sqrt{3}G_{yz}\left(x-y\right)\left(x+y\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_u, 1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{7\sqrt{2}G_u xz\left(x^2-3y^2\right)}{8} - \frac{\sqrt{6}G_v xz\left(3x^2+3y^2-4z^2\right)}{24} + \frac{\sqrt{6}G_{xy}yz\left(3x^2+3y^2-4z^2\right)}{24}$$
$$+ \frac{\sqrt{6}G_{xz}\left(5x^4-21x^2y^2-9x^2z^2+2y^4+9y^2z^2\right)}{24} + \frac{\sqrt{6}G_{yz}xy\left(11x^2-17y^2+18z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_u, 2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{7\sqrt{2}G_u yz\left(3x^2-y^2\right)}{8} + \frac{\sqrt{6}G_v yz\left(3x^2+3y^2-4z^2\right)}{24} + \frac{\sqrt{6}G_{xy}xz\left(3x^2+3y^2-4z^2\right)}{24}$$
$$- \frac{\sqrt{6}G_{xz}xy\left(17x^2-11y^2-18z^2\right)}{24} + \frac{\sqrt{6}G_{yz}\left(2x^4-21x^2y^2+9x^2z^2+5y^4-9y^2z^2\right)}{24}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](E_{1u})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$-\frac{\sqrt{15}G_u yz}{5} + \frac{3\sqrt{5}G_v yz}{10} - \frac{3\sqrt{5}G_{xy}xz}{10} + \frac{3\sqrt{5}G_{xz}xy}{10} - \frac{\sqrt{5}G_{yz}\left(5x^2-y^2-4z^2\right)}{20}$$

$$\frac{\sqrt{15}G_u xz}{5} + \frac{3\sqrt{5}G_v xz}{10} + \frac{3\sqrt{5}G_{xy}yz}{10} - \frac{\sqrt{5}G_{xz}\left(x^2-5y^2+4z^2\right)}{20} - \frac{3\sqrt{5}G_{yz}xy}{10}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](E_{1u})$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](E_{1u})$
** symmetry

$$-\frac{\sqrt{6}x^3}{4} - \frac{\sqrt{6}xy^2}{4} + \sqrt{6}xz^2$$

$$-\frac{\sqrt{6}x^2y}{4} - \frac{\sqrt{6}y^3}{4} + \sqrt{6}yz^2$$

** expression

$$\frac{\sqrt{30}G_u yz\left(3x^2+3y^2-4z^2\right)}{24} + \frac{\sqrt{10}G_v yz\left(27x^2-y^2-8z^2\right)}{24} - \frac{\sqrt{10}G_{xy}xz\left(13x^2-15y^2-8z^2\right)}{24}$$
$$- \frac{\sqrt{10}G_{xz}xy\left(x^2+y^2-6z^2\right)}{24} + \frac{\sqrt{10}G_{yz}\left(2x^4+3x^2y^2-15x^2z^2+y^4-9y^2z^2+4z^4\right)}{24}$$

$$-\frac{\sqrt{30}G_u xz\left(3x^2+3y^2-4z^2\right)}{24}-\frac{\sqrt{10}G_v xz\left(x^2-27y^2+8z^2\right)}{24}-\frac{\sqrt{10}G_{xy}yz\left(15x^2-13y^2+8z^2\right)}{24}$$
$$-\frac{\sqrt{10}G_{xz}\left(x^4+3x^2y^2-9x^2z^2+2y^4-15y^2z^2+4z^4\right)}{24}+\frac{\sqrt{10}G_{yz}xy\left(x^2+y^2-6z^2\right)}{24}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](E_{2u})$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](E_{2u})$
** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{6}G_u\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{2}G_v\left(x^2+y^2-2z^2\right)}{4}$$

$$-\frac{\sqrt{6}G_u xy}{2}-\frac{\sqrt{2}G_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](E_{2u})$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](E_{2u})$
** symmetry

$$\sqrt{15}xyz$$

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{3}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{6}-\frac{G_v\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{6}$$
$$-\frac{7G_{xy}xy\left(x-y\right)\left(x+y\right)}{6}+\frac{7G_{xz}xz\left(x-z\right)\left(x+z\right)}{6}-\frac{7G_{yz}yz\left(y-z\right)\left(y+z\right)}{6}$$

$$\frac{\sqrt{3}G_u xy\left(x^2+y^2-6z^2\right)}{3}+\frac{7G_v xy\left(x-y\right)\left(x+y\right)}{6}-\frac{G_{xy}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{12}$$
$$-\frac{7G_{xz}yz\left(3x^2+y^2-2z^2\right)}{12}-\frac{7G_{yz}xz\left(x^2+3y^2-2z^2\right)}{12}$$